



AI-Powered Sentiment Tracking and Support System for Emotional Well-Being



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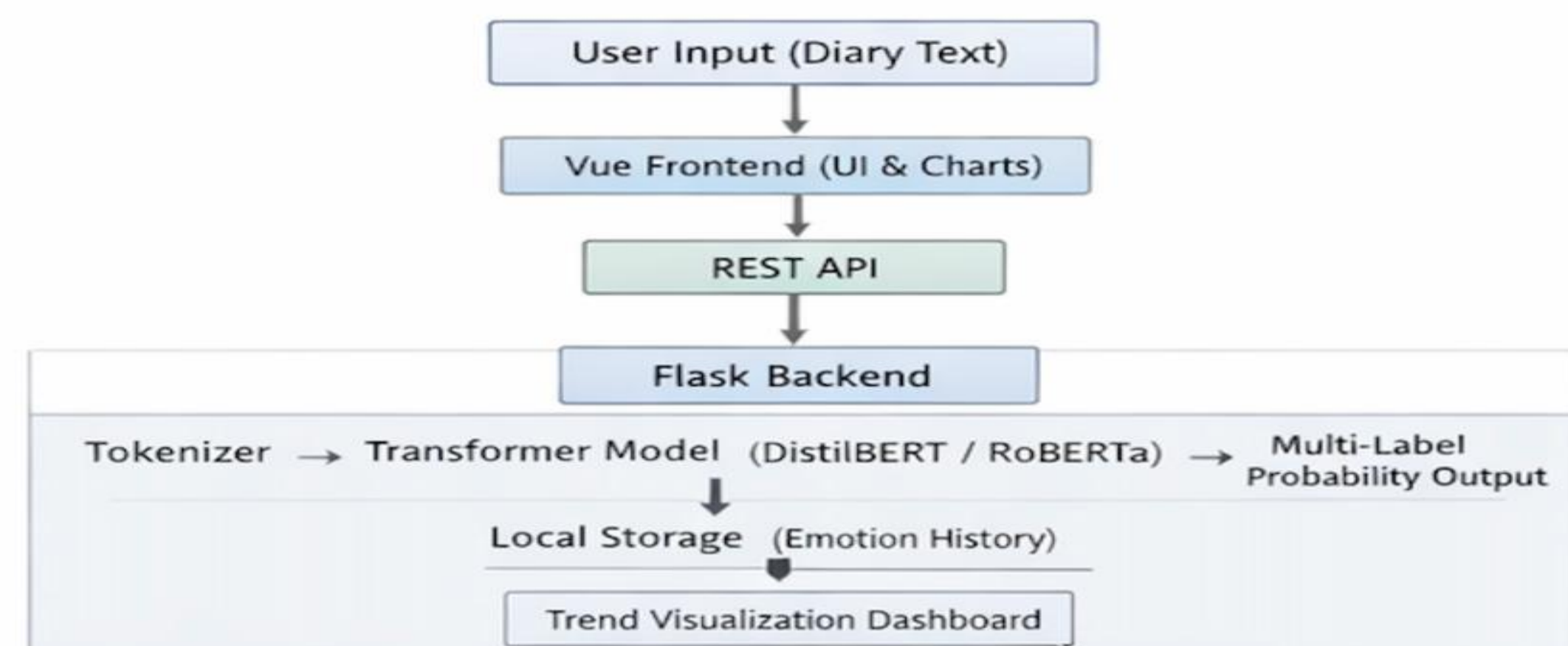
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Introduction

University students increasingly use digital mental health tools for support and self-reflection. However, traditional sentiment analysis usually reduces emotion to only three categories—positive, negative, or neutral—which is too limited for mental well-being monitoring [1][2]. Real emotional states are often mixed and more nuanced than a single label can represent. Many AI-based mental health tools also rely on cloud processing, which creates privacy risks for sensitive personal text. To address both the limitations of coarse sentiment classification and the privacy concerns of cloud-based systems, this project develops a “Privacy First” system that uses local inference for multi-label emotion tracking [3].

Methods and Materials

Data from publicly available emotion datasets were combined and preprocessed into a unified training set. Label harmonization was applied to map different taxonomic structures into a shared multi-label framework. Transformer-based models, including DistilBERT and RoBERTa, were fine-tuned for emotion classification. DistilBERT was selected as the primary model for efficient local inference, while RoBERTa-large served as a higher-capacity reference model [4]. The system used a multi-label classification approach to identify multiple emotional labels from free-form journal entries. This design better reflects the mixed and nuanced emotions often expressed in reflective writing.



Results

Quantitative Performance: The model showed stable learning and reliable classification performance. Training loss decreased from about 0.30 to 0.10, while validation loss dropped from around 0.55 to 0.26. The confusion matrix also showed strong category-level results, with scores of 1.00 for Joy and Surprise, 0.89 for Anger, Fear, and Love, and 0.80 for Sadness. **Qualitative Usability:** User feedback indicated that the system was generally understandable and helpful for self-reflection. Most participants reported that the dashboard made emotion patterns easier to interpret over time, and the questionnaire results suggested positive usability and satisfaction with the interface. **System Responsiveness:** The local-first architecture enabled fast interactive feedback during journal writing while keeping sensitive text on the user’s device. This design improved privacy protection and made the system practical for reflective daily use.

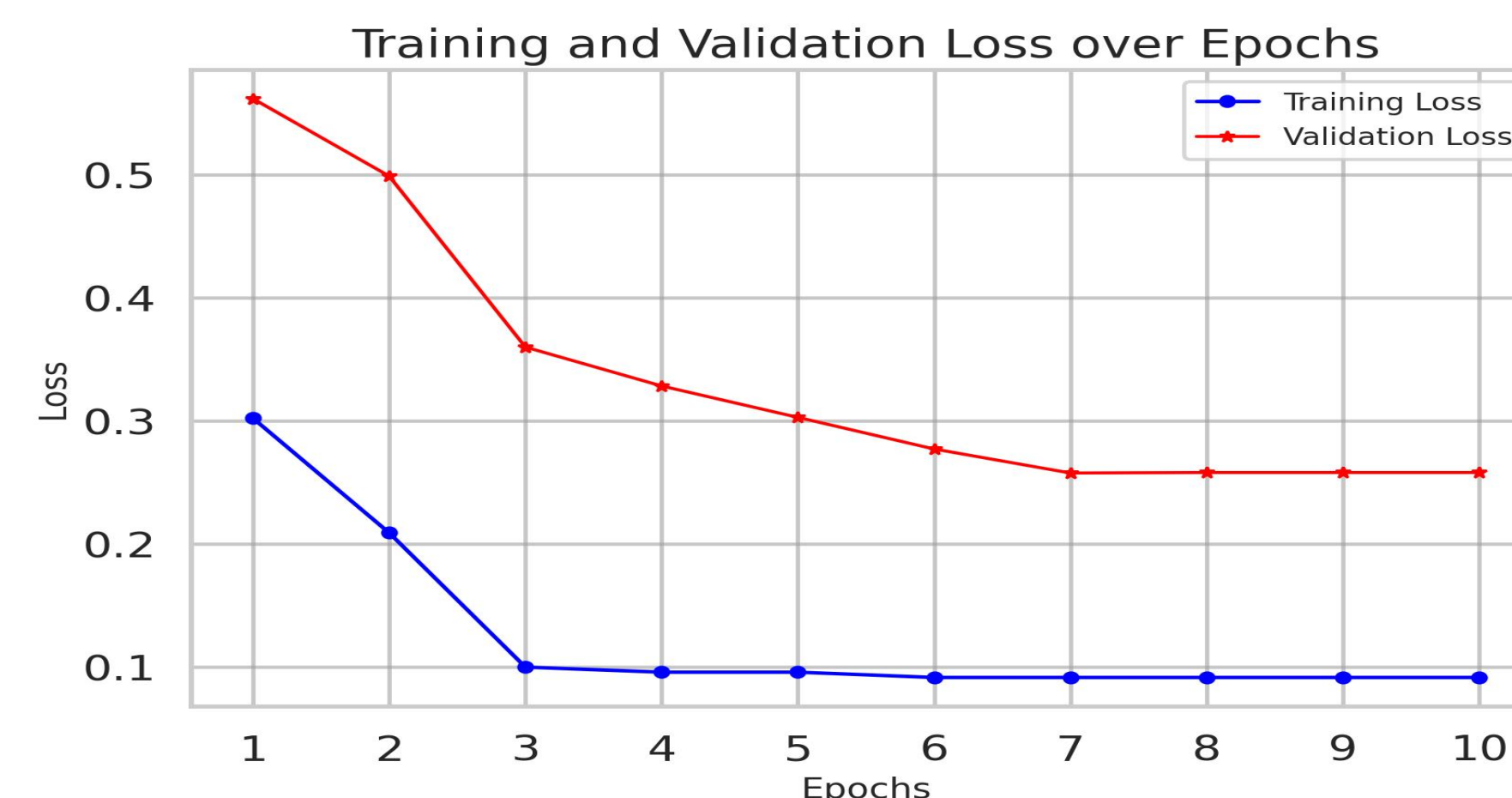


Figure 4.2: Surrogate loss across staged rule-based model refinement

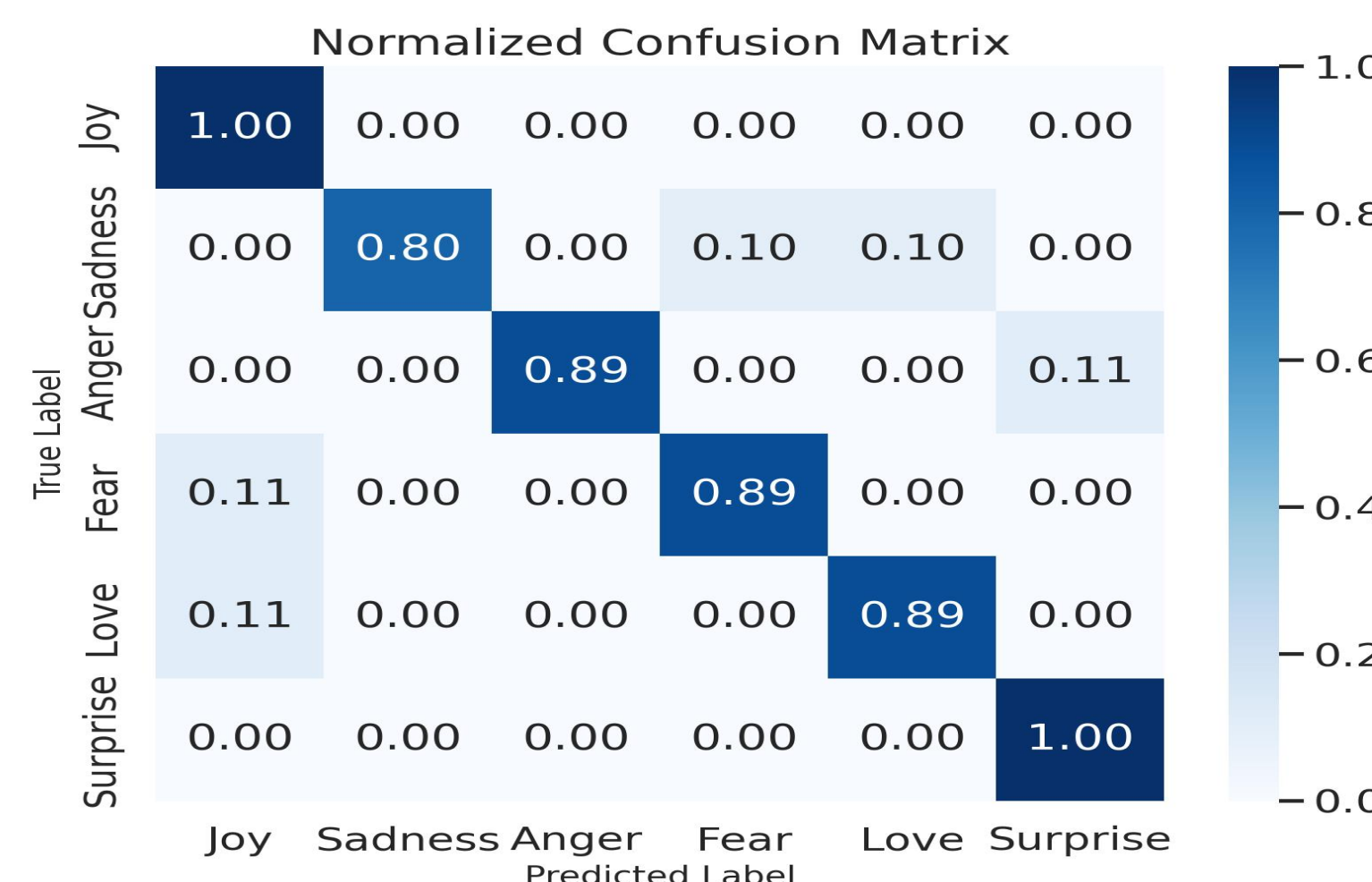


Figure 4.3: Final-model confusion matrix on the thesis-aligned benchmark

Conclusions

This project illustrates how to create an artificial intelligence (AI) application for tracking emotion using Natural Language Processing (NLP) that will keep a user's emotional state private. Using transformer-based architectures as well as interactive visualization systems, this project allows users to evaluate their feelings through journaling. Results show that lightweight transformer architecture can produce accurate emotion classification results and efficient operation. This project serves as a proof of concept for how the combination of machine learning technologies with human-centered approaches can improve the emotional self-awareness. Future development efforts will be directed at improving the robustness of the emotion classification models; developing emotion datasets; and developing adaptively personalized features for individual users.

References

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